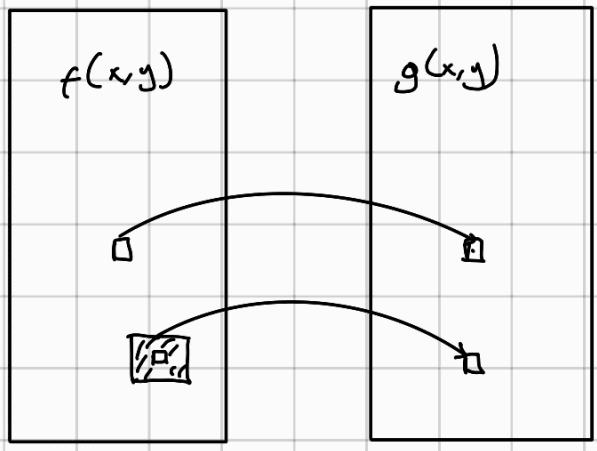
Image depth : Bir piksel ifade etmek için kullanılan bit sayısı

Image Enhancement: Process an image to obtain an image that is more suitable than the original image for a specific application

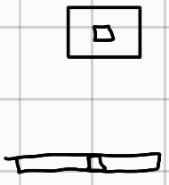
$f(x,y)$

$g(x,y)$

$g(x,y) = T[f(x,y)]$



kare olması gerekmez

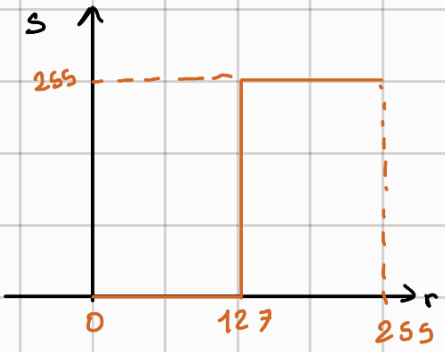


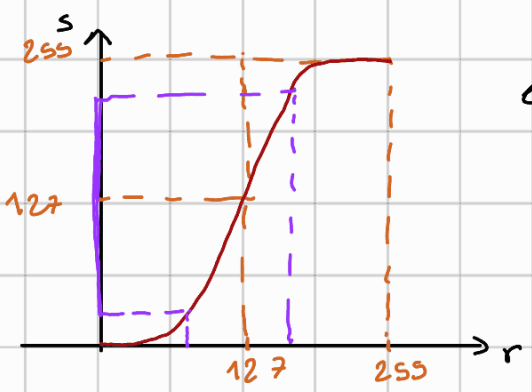
Operation on |x| neighborhood

- $g(x,y)$ depends only on the value of f at (x,y)

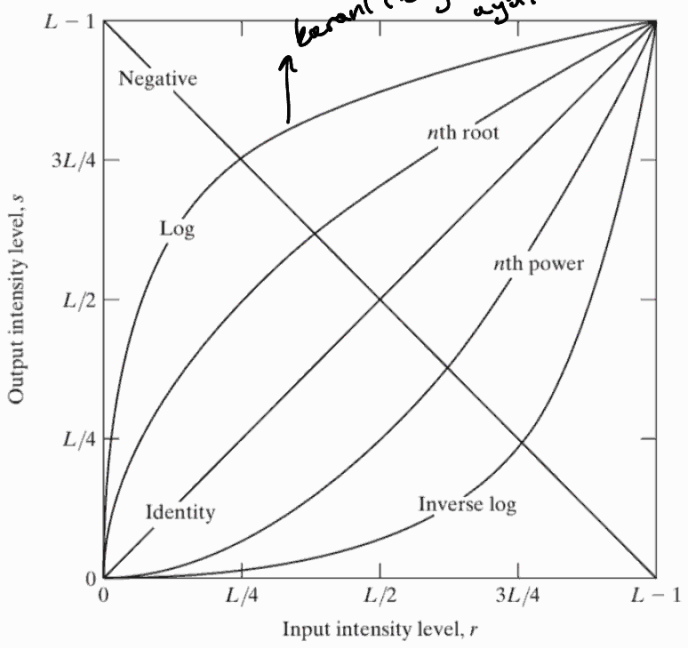
$g(x,y) = T[f(x,y)]$

$s = T[r]$





contrast stretching function

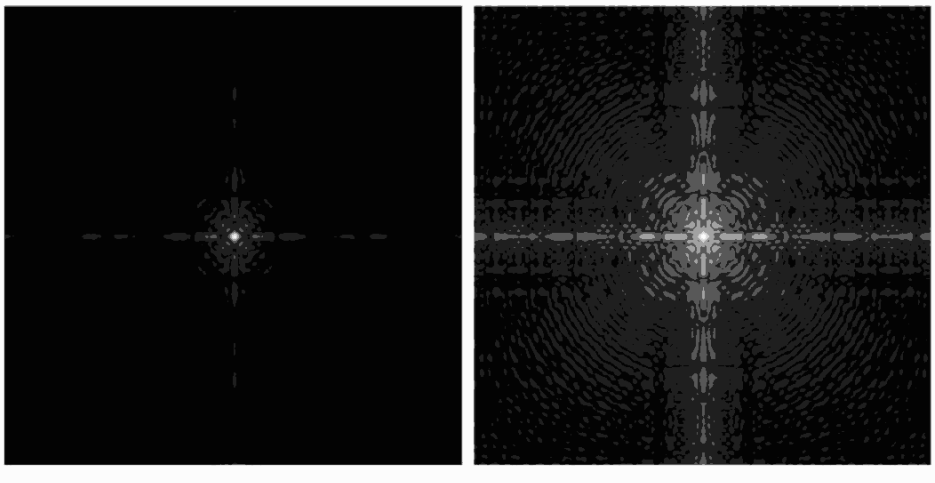


karanlık görüntüyü aydınlatır (karanlık bölgeyi daha iyi görmeni sağlar)

logaritmayı yüksek bileşenleri bastırmak için kullanırız.

a b

FIGURE 3.5
(a) Fourier spectrum.
(b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.



3.2.3 Power-Law (Gamma) Transformations

Power-law transformations have the basic form

$$s = cr^{\gamma} \tag{3.2-3}$$

where c and γ are positive constants. Sometimes Eq. (3.2-3) is written as $s = c(r + \varepsilon)^{\gamma}$ to account for an offset (that is, a measurable output when the input is zero). However, offsets typically are an issue of display calibration

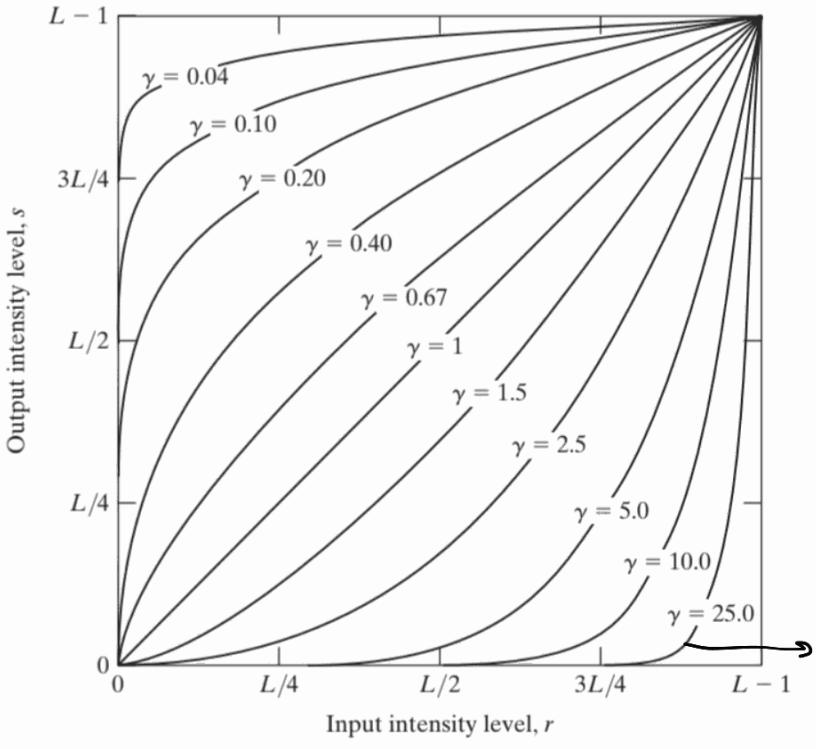
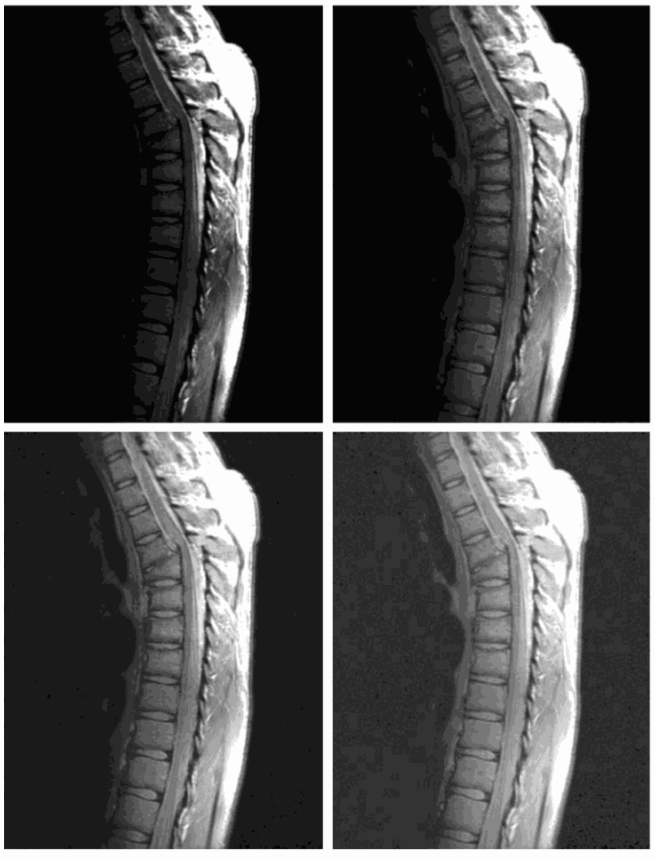


FIGURE 3.6 Plots of the equation $s = cr^{\gamma}$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.

çok aydınlık yerleri

karartarak ve gayarak detaylandırır



a b
c d

FIGURE 3.8 (a) Magnetic resonance image (MRI) of a fractured human spine. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6$, 0.4, and 0.3, respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

a b
c d



FIGURE 3.9
(a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0$, 4.0, and 5.0, respectively. (Original image for this example courtesy of NASA.)

a b
c d

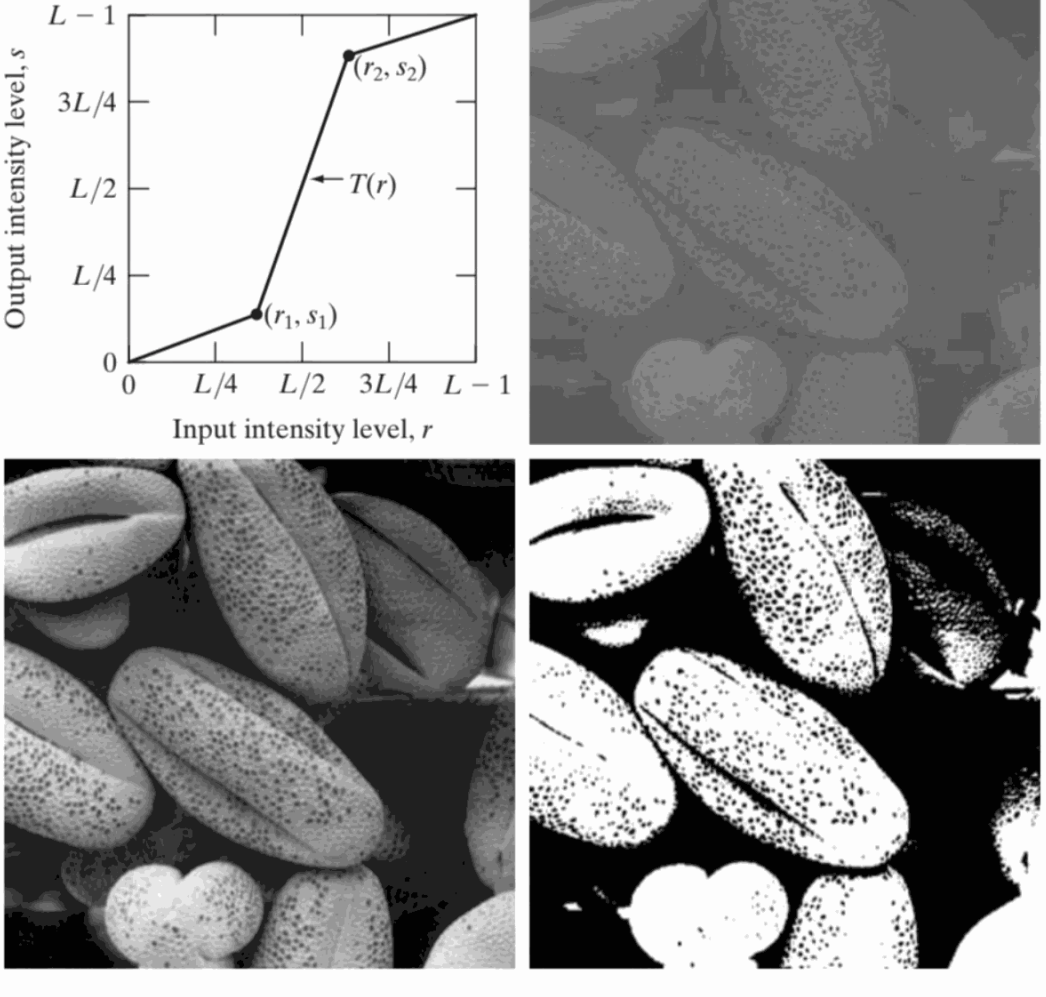
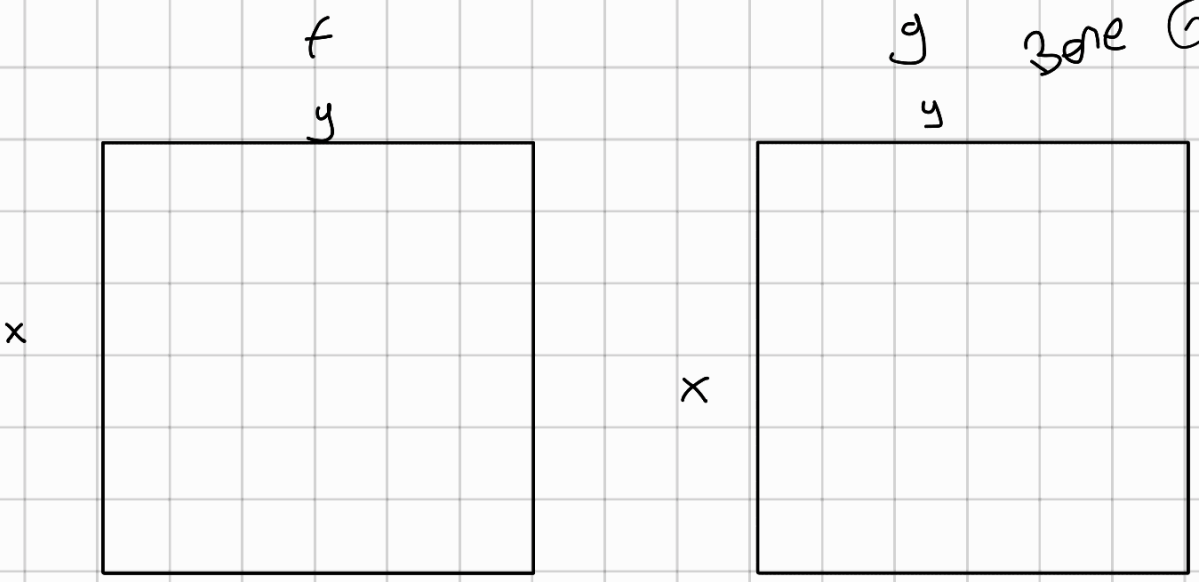


FIGURE 3.10
Contrast stretching. (a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Hasan El-Gaib




```
for x=0 to M-1
    for y=0 to N-1
        g(x,y)=T[f(x,y)]
```

r

300

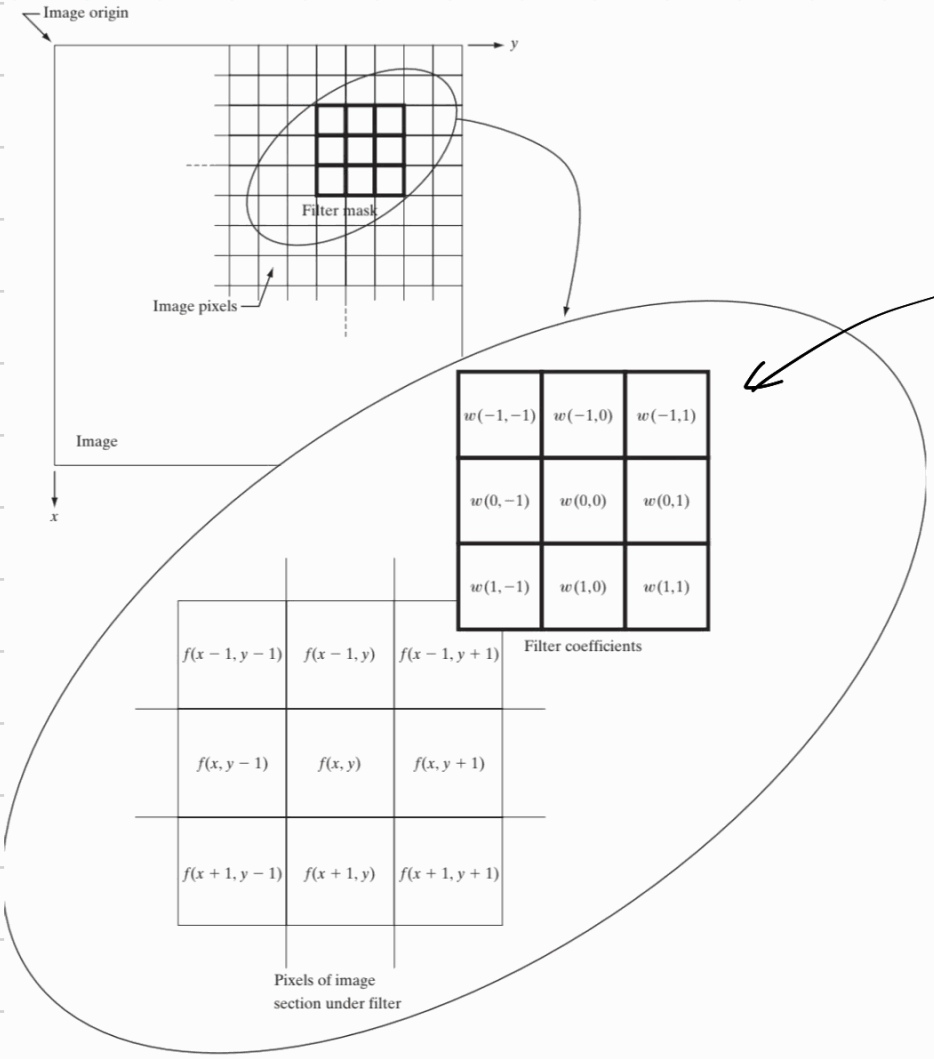
-100 (min)

r - min = r + 100

400

0

0 (varsa) { (r-min)/max . 255 } 255 / 0



bu korelasyon işlemi konvülasyondan farklı ters alma yoktur. Simetrik olursa korelasyon ve konvülasyonun farkı kalmaz

konvulasyon

x[n]		1	2	3	4	3	6	7	8 9
h[n]		5	4	3					
	3	4	(5)						
x[n]*h[n]		5	14	26					

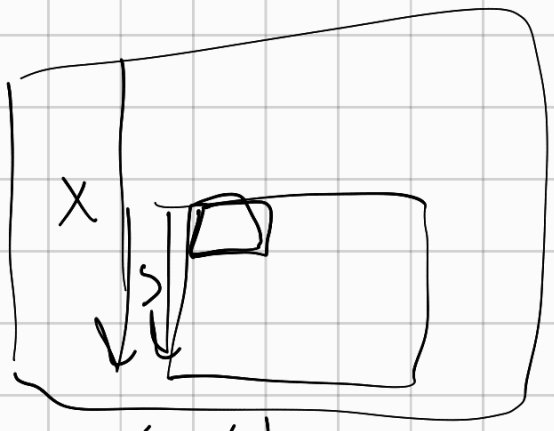
Görüntün dışında kalan pikselleri 0 olarak kabul edin projede.

Mesela bu maskeyi düşünelim

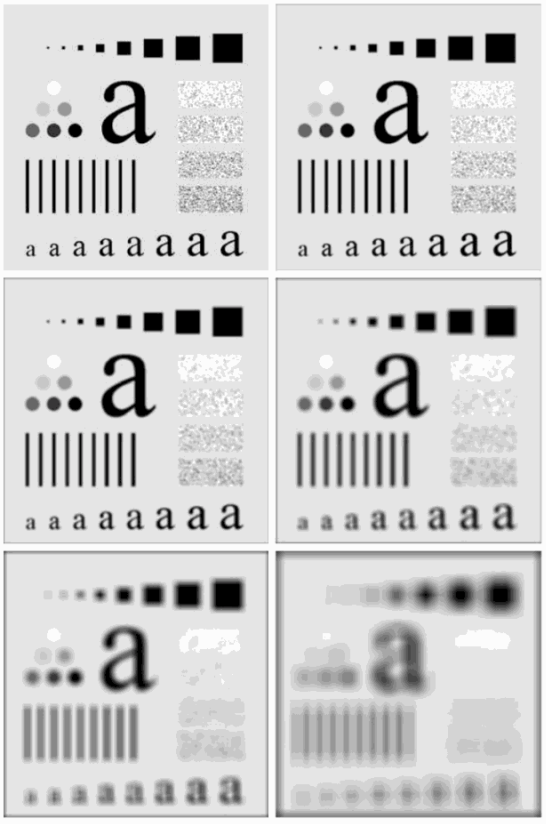
s \ t			
	1	1	1
	1	1	1
	1	1	1

```
for x=0 to M-1
    for y=0 to N-1
        g(x,y)=T[f(x,y)]
```

```
g(x,y)=0
for s=-1 to +1
    for t=-1 to +1
        g(x,y)=g(x,y)+f(x+s,y+t)
    g(x,y)=g(x,y)/9
    if x+s<0 || x+s>M-1
        || y+t<0 || y+t>N-1 yapma
    else:
```

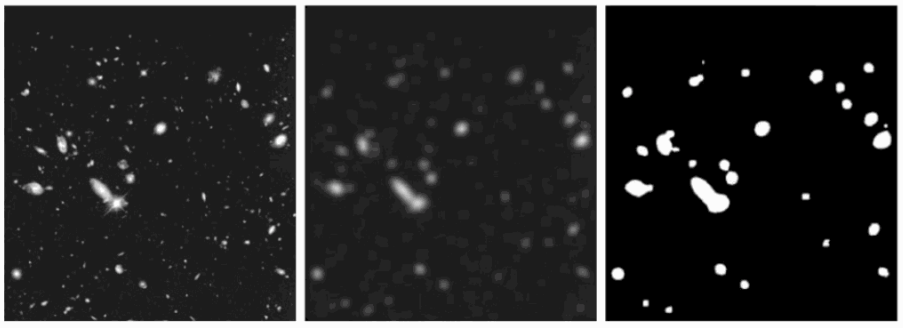


$$g(x,y)=\sum_{s=-1}^{1}\sum_{t=-1}^{1} f(x+s,y+s)w(s,t)$$



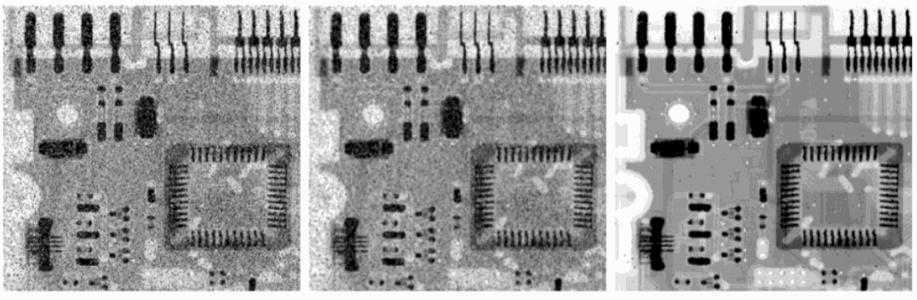
a b
c d
e f

IGURE 3.33 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing vith square averaging filter masks of sizes m = 3, 5, 9, 15, and 35, respectively. The black quares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders re 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in ncrements of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels vide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 ixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% lack in increments of 20%. The background of the image is 10% black. The noisy ectangles are of size 50 × 120 pixels.



a b c

FIGURE 3.34 (a) Image of size 528 × 485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15 × 15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)



a b c

FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3 × 3 averaging mask. (c) Noise reduction with a 3 × 3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

pencere içindeki değerleri sıralıyorum
küçükten büyüğe doğru
ve ortanca değeri almış gibi

En küçüğü alsa min, en büyüğü alsa max, kninci

alsa k'ıncı filtre

Başka filtreye bakalım

1	2	1
2	4	2
1	2	1

ortadaki değerin ağırlığı daha fazla

Bölmeden girişe uygulandığını düşünelim

255×16 oldu

Bu yüzden filtre başta 16'ya bölünmek

maskenin toplamı bir olmalıdır.

Sharpening Spatial Filters

keskinleştirme detayları daha fazla görünür hale getirmek için

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$ Laplacian

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

0	1	0
1	-4	1
0	1	0

min -4×255

max 4×255

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

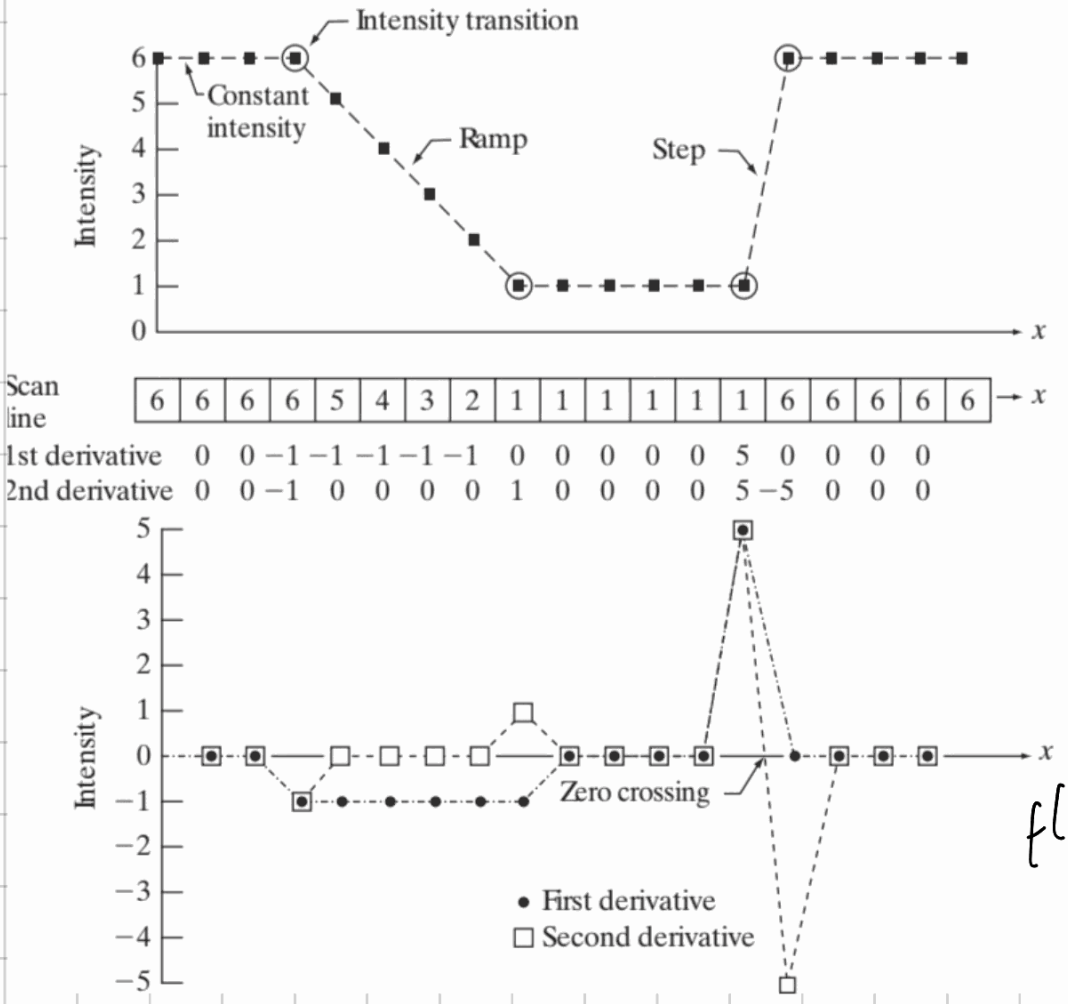
0	0	0
0	1	0
0	0	0

$-$

0	1	0
1	-4	1
0	1	0

$=$

0	-1	0
-1	5	-1
0	-1	0



a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.